



## Building A Simple Durable Contact Microphone

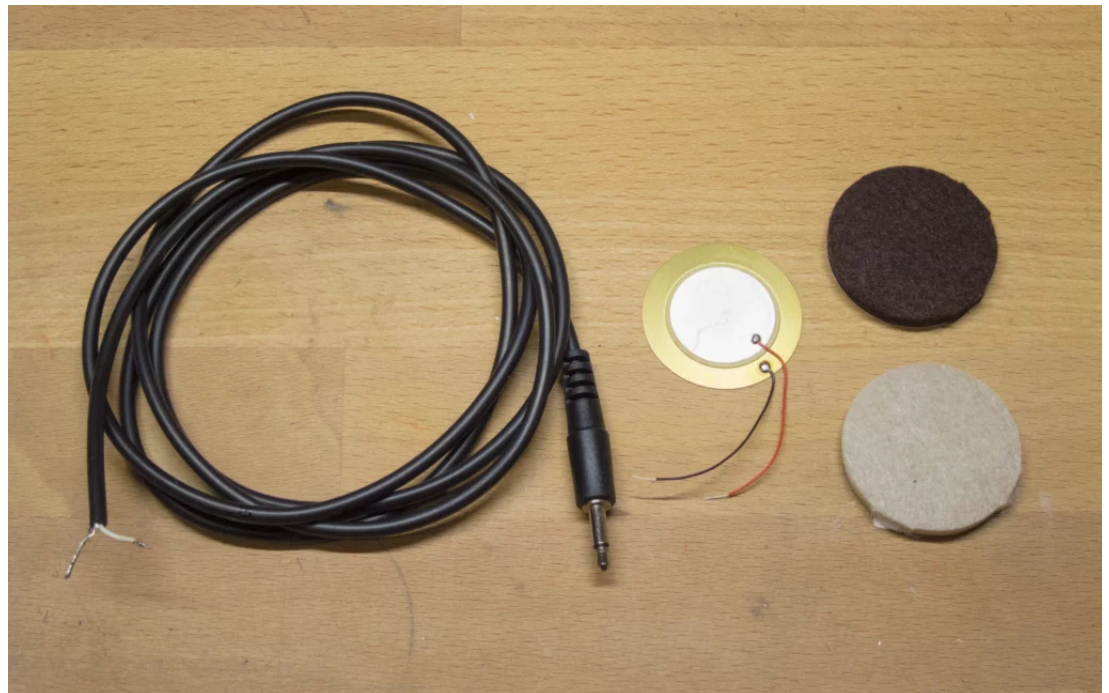
(This page is part of a series. See the rest of my [contact microphone pages](#) here.)

This is my step-by-step guide to building simple DIY contact microphones. There are many other ways to do it, but my goals with this method are **simplicity** and **durability**.

### What you need:

#### Parts:

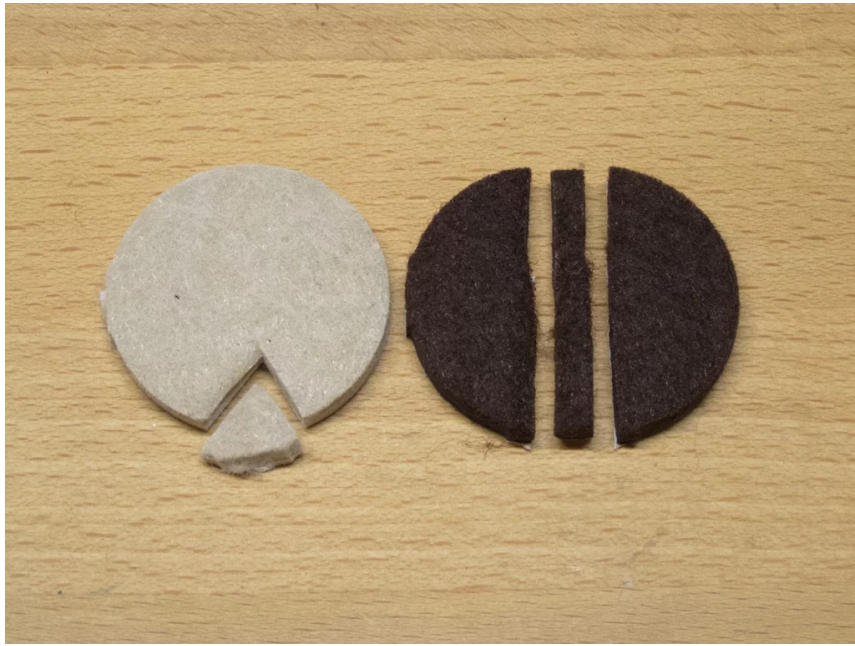
- **Shielded audio cable & plug**  
(Hint: Choose your plug based on what preamp you plan to use (usually 1/4" or 1/8"). Save some soldering by buying a cable with the plug already molded into one end, or buy a pre-made cable and cut off one end to reveal the wires. )
- **35mm piezo "bender" disc** Hint: Get one with wires attached so you don't vaporize the fragile plating on the piezo ceramic. I use Murata 7BB-35-3L0. Mail-order from [Mouser](#) or buy in NYC from [Tinkersphere](#). (These are intended as tiny resonant speakers, but we will misuse them as microphones.)
- **Two felt discs** intended to protect floors from furniture legs. 1<sup>1/2</sup> inch size will fit 35mm piezos perfectly. (I used two different colors here, but color doesn't matter.)



#### Tools:

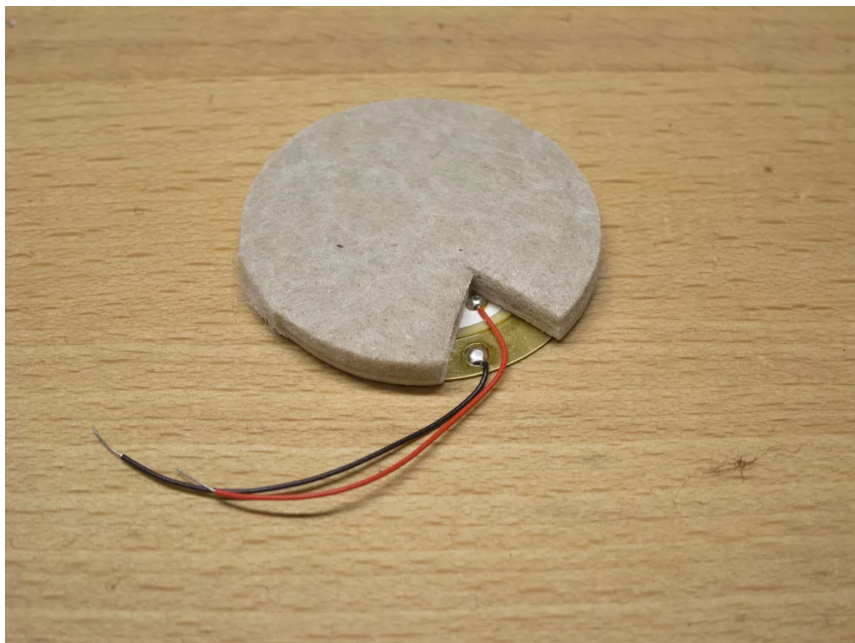
- Soldering iron & solder
- Hot Glue Gun
- Knife / Scissors / Wire-Cutters

# Let's Build!



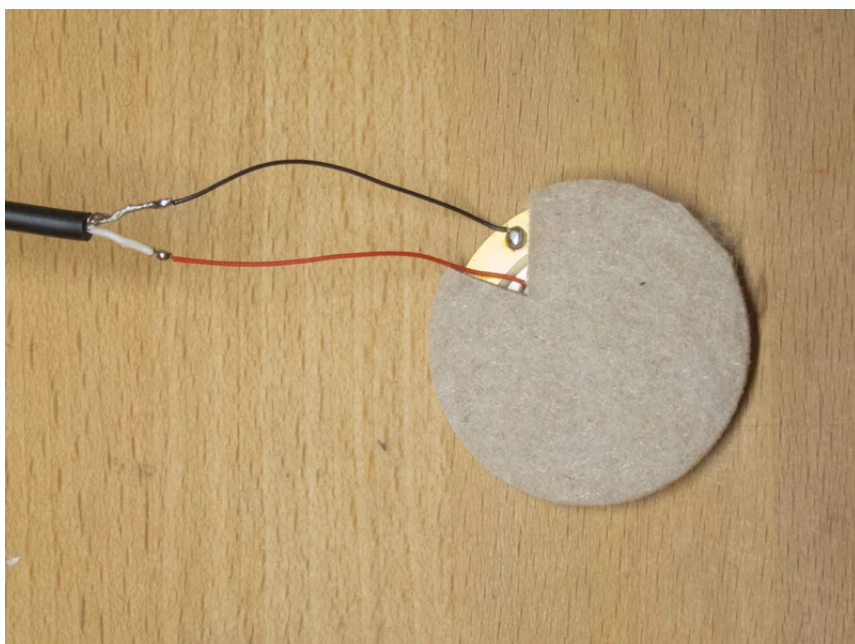
## Step 1

Cut a little pizza slice from one disc (and save it for later). Remove a narrow canyon from the other (no need to save).



## Step 2

Stick the pizza to the piezo disc. The missing slice should reveal the solder points and allow the wires to escape.



## Step 3

Solder the cable to the piezo wires:  
Center of piezo (red wire) = center conductor of cable.  
Rim of piezo (black wire) = shield of cable.

**Pro Tip:** You may want to hot-glue the cable on top first (see the next step). That will keep things from moving around while you solder.

## Step 4

Hot-glue the cable to the top of the pizza. Groom the wires so they run along the cable, and return the missing slice. (It helps keep the wires from accidentally touching.)

## Step 5

Stick the canyon on top of the pizza, so the cable runs down the middle of the canyon.

## Step 6

Fill the canyon with hot glue. It's a good idea to seal the edges and build up some glue around the cable exit as a strain-relief.

## You're done!

Use a little dot of reusable poster adhesive to stick the mic onto things. (The exposed brass side should always touch the object.)

There are other methods of attaching the mic (clamps, heavy rocks, magnets). The general rule of thumb is more pressure = more loudness & bass.

## Example

Here's a little improvisation exploring the many voices of the humble pot lid (via a buffer preamp):

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## You Should Use A Preamp!

Contact mics have extremely high electrical impedance, which is not well-matched to the mic inputs on recorders and mixers. You'll get OK signal levels but very little bass, and an exaggerated "screechy" treble. The solution is to add a buffer preamplifier between the mic and the recorder. Read more about it [here](#).

# Mods and Upgrades

These simple mics have a very high output, but they possess a **resonant mid-range “honk”** that tends to color all of your recordings. The felt discs will damp some of this natural resonance, but it’s still there. Ways to reduce the honk:

- Cut the edges off the piezo with metal shears (careful not to crack too much of the ceramic). This may increase the resonant frequency out of the mid-range.
- Glue a plastic or metal disc to the bottom (or try other hard materials). This dramatically flattens the frequency response at the expense of sensitivity. (No free lunch folks!) I show this here. (ADD LINK HERE)
- Use a different transducer with less inherent resonance (see options in the Choosing a Contact Mic page).

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The recording above is a comparison of three mics clamped to a piece of metal as I draw circles with a pencil on the surface:

1. 35mm bender as shown on this page: *Severely lacking in high frequency detail*
2. Upgraded with acrylic disc attached: *Biased toward high frequencies at the expense of lower midrange / bass.*
3. Humidifier disc: *Pretty balanced, although the output is lower than the benders.*

Your contact mic will probably pick up **stray electromagnetic fields**. To prevent it, insulate the top ceramic of your piezo disc with electrical tape. Then add a layer of copper foil tape on top, making sure that the foil touches the brass outer ring. (Solder the foil to the brass in a few places.) This will trap the ghosts before they get into your signal.

Dip the finished mic in Plasti-Dip (or similar liquid rubber coating) to **waterproof it**. The soft rubber has less effect on the frequency response than I expected, but it will reduce the treble. (It probably won't make a good hydrophone. You need slightly different construction for efficient underwater recording.)

Interested in recording bridges, buses, or light poles? Embed a **strong magnet** inside the mic so it sticks to ferrous surfaces without clamps.

## Related Posts:

[Choosing a Contact Mic For Field Recording](#)

[3D-Printed DIY Contact Microphone](#)

[Alex Rice Piezo Preamplifier](#)

[Zach's Recorder Recommendations](#)

[Choosing a Hydrophone For Field Recording](#)

[Sound Art Resources](#)

[Contact Microphone Pages](#)

[Smith Victor 765: Taming The Glass](#)

[Repair Amaran P60c LED Panel](#)